

4 Route Values and Organizational Sorting

This section composes the commercial-value kernel with the internal and entrusted manufacturing technologies defined above. It treats the qualified manufacturing-service price p_m as fixed and conjectured. Market consistency for p_m^* is imposed in the capacity-market section below.

4.1 Route values

For an internally feasible project, the internal route value is

$$W_i^I(q, m) = s(q)R(q, c_I(m, k_i)) - F_I(m, k_i). \quad (1)$$

If $k_i < \underline{k}(m)$, then $F_I = +\infty$ and $W_i^I = -\infty$, so internal production is unavailable. This is an extended-value convention; we do not evaluate the undefined internal marginal cost below feasibility.

At a conjectured CMO capacity price p_m , retained entrusted manufacturing has value

$$\begin{aligned} W_i^E(q, m; M, p_m) &= s(q)R(q, c_E(m)) - F_E(m) - p_m b(m) \\ &\quad - \mu_E - \tau_E(M). \end{aligned} \quad (2)$$

The developer remains the authorization holder under E ; the qualified external producer manufactures. Each route cost is subtracted exactly once. In particular, $p_m b(m)$ is not contained in $c_E(m)$.

Let the noncore transfer/out-license outside value be a finite continuous function $T(q, m)$, and normalize abandonment or indefinite delay to zero:

$$W^T(q, m) = T(q, m), \quad W^A = 0. \quad (3)$$

No separate transfer market is added.

4.2 Deterministic route choice

The optimized value of a planning-stage project is

$$W_i(q, m; M, p_m) = \max \left\{ W_i^I(q, m), W_i^E(q, m; M, p_m), W^T(q, m), W^A \right\}. \quad (4)$$

The route choice is

$$r_i^*(q, m; M, p_m) = \arg \max_{r \in \{I, E, T, A\}} \begin{cases} W_i^I(q, m), & r = I, \\ W_i^E(q, m; M, p_m), & r = E, \\ W^T(q, m), & r = T, \\ W^A, & r = A. \end{cases} \quad (5)$$

Alongside continuous heterogeneity, impose the no-tie regularity restriction: for each fixed candidate price and regime, maximizing ties have zero project probability for almost every developer. Continuity alone does not imply this restriction. A fixed measurable selection resolves null-set ties. Thus the choice is single valued almost surely. This is a deterministic comparison; it generates no probabilistic route share.

When $M = 0$, $\tau_E(0) = +\infty$ implies $W_i^E = -\infty$. At fixed p_m , the binary reform comparison is therefore

$$\begin{aligned} &W_i(q, m; 1, p_m) - W_i(q, m; 0, p_m) \\ &= \max \left\{ W_i^E(q, m; 1, p_m), W_i^I(q, m), W^T(q, m), 0 \right\} \\ &\quad - \max \left\{ W_i^I(q, m), W^T(q, m), 0 \right\} \geq 0. \end{aligned} \quad (6)$$

The value gain is exactly zero whenever the newly feasible entrusted value fails to exceed the best pre-reform route. The route is unchanged when the inequality is strict; equality can affect only a tie selection, a null event under the stated regularity restriction.

4.3 Internal-entrusted value gap

For fixed q, m, M, p_m with finite $\tau_E(M)$ and internal feasibility, define

$$\begin{aligned}\Delta_{IE}(k_i; q, m, M, p_m) &= W_i^I(q, m) - W_i^E(q, m; M, p_m) \\ &= s(q)[R(q, c_I(m, k_i)) - R(q, c_E(m))] - F_I(m, k_i) \\ &\quad + F_E(m) + p_m b(m) + \mu_E + \tau_E(M).\end{aligned}\tag{7}$$

On the interior of the internally feasible domain, only the internal technology terms vary with k_i . Holding q, m, M, p_m fixed,

$$\frac{\partial \Delta_{IE}}{\partial k_i} = s(q)R_c(q, c_I(m, k_i))c_{I,k}(m, k_i) - F_{I,k}(m, k_i) > 0.\tag{8}$$

The first term is weakly positive because $s(q) \geq 0$, $R_c < 0$, and $c_{I,k} < 0$; the second is strictly positive because $F_{I,k} < 0$. Hence the entire derivative is strictly positive, including the boundary case $s(q) = 0$.

4.4 Conditional organizational cutoff

Consider a finite entrusted barrier and suppose that the continuous value gap satisfies the endpoint crossing conditions

$$\lim_{k \downarrow \underline{k}(m)} \Delta_{IE}(k) < 0, \quad \lim_{k \rightarrow \infty} \Delta_{IE}(k) > 0.$$

Continuity gives existence of a root, and (8) gives strict monotonicity and therefore uniqueness. Denote the unique root by

$$\Delta_{IE}(k^*(q, m; p_m, M); q, m, M, p_m) = 0.\tag{9}$$

Conditional on both I and E dominating transfer and abandonment,

$$k_i < k^* \Rightarrow r_i^* = E, \quad k_i > k^* \Rightarrow r_i^* = I.\tag{10}$$

If T or A dominates, the I/E cutoff does not determine the realized route.

For a finite wedge, let the arguments τ_E and p_m vary locally while holding q, m fixed. Since

$$\frac{\partial \Delta_{IE}}{\partial \tau_E} = 1, \quad \frac{\partial \Delta_{IE}}{\partial p_m} = b(m) > 0,$$

the implicit-function theorem yields

$$\frac{\partial k^*}{\partial \tau_E} = -\frac{1}{\Delta_{IE,k}(k^*)} < 0, \quad \frac{\partial k^*}{\partial p_m} = -\frac{b(m)}{\Delta_{IE,k}(k^*)} < 0.\tag{11}$$

A lower finite institutional barrier raises k^* , expanding the entrusted-production region $k_i < k^*$. A higher fixed CMO price lowers k^* , contracting that region. The price derivative is a conditional comparative static, not an equilibrium derivative through p_m^* .

4.5 Organizational sorting result

Suppose the commercial-value and technology restrictions above hold, the entrusted barrier is finite, the endpoint crossing conditions hold, and ties have measure zero. Then the internal-minus-entrusted value gap is strictly increasing in developer manufacturing capability, a unique finite I/E cutoff exists, and the conditional sorting and implicit derivatives in (10)–(11) follow. The proposition is conditional: it does not displace the valid transfer and abandonment outside options.

4.6 Zero-effect and non-tautology boundary

The route comparison is not tautological. Internal value depends on developer capability through c_I and F_I ; entrusted value instead uses qualified-external technology, a physical capacity requirement, a retained holder burden, and the institutional wedge. MAH can nevertheless have zero effect when E remains below the pre-reform maximum, including when T or A still dominates after reform. Failure of the finite I/E crossing condition alone does not imply zero reform gain. No probabilistic route share is used.